

$$\boxed{2.17} \quad \text{Cond}(A) \geq 1, \quad \forall A.$$

Sol:

$$\text{Cond}(A) = \|A\| \cdot \|A^{-1}\|, \quad \text{unde } \|\cdot\| \text{ este o normă matricială}$$

$$I = A \cdot A^{-1}$$

$$\Rightarrow \underbrace{1} = \underbrace{\|I\|} = \|A \cdot A^{-1}\| \leq \|A\| \cdot \|A^{-1}\| = \underline{\text{Cond}(A)}$$

$$\Rightarrow \text{Cond}(A) \geq 1, \quad \text{pt } \forall A \text{ neregular.} \quad \blacksquare$$